

Audit Checker Rebuild

Technical brief for the developer building our audit + report generator. Read in full before starting. Questions go in our shared doc.

BONUS PAY

+\$100 for implementing the dynamic checker changes described in Section 3.

+\$250 for implementing both the Lite Report and Full Report with shared state (so the same competitor appears in both for the same lead).

Total bonus: \$350 on top of the current scope.

What's changing

We are restructuring the audit into two outputs from one checker:

1. Lite Report — small JSON payload used to write the cold email that goes to the lead. Cheap. Runs on every prospect in our outreach list. Not shown to the lead directly.

2. Full Report — the 6-page PDF we send to leads who reply "send." Expensive. Runs only on leads who ask for the report.

Both outputs must reference the **same competitor** for the same lead, with consistent data. See Section 4 for the persistence requirement.

SECTION 1

Finding the Right Competitor

The competitor we compare against is the most important piece of input for the whole report. Get this wrong and nothing else works.

Selection rules

When the checker receives a lead, it must find a competitor that meets all of these criteria:

- › Same city or metro as the lead. Not nationwide brands. Example: lead in Toledo, OH → competitor must also be a Toledo, OH business the lead would actually recognize.
- › Same vertical. Plumber vs plumber. Roofer vs roofer.
- › Meaningfully ahead of the lead in Google Maps. Not just one spot ahead — a clear gap.

Gap rules (lead vs competitor positions)

The selected competitor must outrank the lead by a real margin. Use these thresholds:

IF LEAD POSITION IS	COMPETITOR POSITION SHOULD BE
4–10 in map pack	Top 1–3
11–20 in local results	Top 1–5
Below top 20	Top 1–10
Top 3 already	Skip — find a different angle (more reviews, better GBP score)

EXAMPLE

Lead: Empire Sewer and Water, Toledo, OH, position #10 in maps for "plumber Toledo OH."
Competitor selected: a Toledo plumber sitting at #3 — clearly ahead, in the same city, recognizable to the lead. Not a national brand.

Required APIs

These are the data sources the checker needs to find and verify the competitor:

API / SERVICE	USED FOR	COST
Google Places API (Place Search + Place Details)	Find the lead's GBP, fetch competitor GBP data, get reviews, rating, photos, hours, categories	~\$0.017 per call

SerpAPI (engine=google_local) OR DataForSEO Local SERP	Scrape map pack rankings for the target keyword in the lead's city. Returns top 20 results with positions.	~\$0.005–0.01 per query
Google PageSpeed Insights API	Site speed scores for both the lead's site and the competitor's site (mobile + desktop, Core Web Vitals).	Free (rate-limited)
Puppeteer or Playwright (self-hosted)	Crawl both sites to detect: embedded review widgets, CTA visibility, trust badges, mobile phone visibility, domain mismatch, service area pages.	Hosting only (~\$10/mo)
Outscraper or DataForSEO Reviews	Bulk pull of reviews + response rates from GBP (faster than calling Places API per review).	~\$0.02 per business

SECTION 2

Two Reports from One Checker

We need **two different outputs** from the same audit logic. The lead data must be consistent between them.

Lite Report (used for cold email writing)

Runs on every lead in our outreach batch (typically 100 per day). Output is a small JSON we pass to Claude to write the cold email. The lead never sees this report directly. Should be **short, cheap, and surface-level**:

- › Lead's GBP rating + review count
- › Lead's position in map pack for primary keyword
- › Selected competitor's name, position, rating, review count
- › The single most striking paradox or gap (1 sentence)
- › Owner first name (if available)

COST TARGET — LITE REPORT

Under **\$0.05 per lead**. At 100 leads/day that's \$5/day or \$150/month. If it costs more, find a cheaper data source.

Full Report (the 6-page PDF)

Runs only when a lead replies "send" to our cold email. Output is the PDF (see reference:

Empire_Audit_NEW_STRUCTURE.pdf). Includes deep site crawl, full GBP analysis, speed tests, competitor side-by-side.

COST TARGET — FULL REPORT

Up to **\$0.50 per report**. Triggered manually or auto-on-positive-reply. Volume is low (10–20% of leads who got the cold email).

SECTION 3

Persistence — The Same Competitor in Both

CRITICAL RULE

When the Full Report runs for a lead, the competitor selected must be **the exact same competitor** referenced in their Lite Report. If a different competitor appears in the Full Report than in the email the lead received, our credibility is destroyed.

How it works

The reports are **not** generated back-to-back. The flow is:

STEP	WHAT HAPPENS	TIMING
1	Lite Report runs for the batch. Data saved per lead.	Day 0
2	Cold email goes out using Lite Report data.	Day 0
3	Lead replies "send" (or doesn't).	Day 0–5
4	Full Report triggered. Reads the saved data first, uses the same competitor, then runs deeper checks.	Day 0–5

Implementation

Store Lite Report output in a database keyed by lead_id (or domain). When Full Report runs:

- 1. Look up the lead's Lite Report record by lead_id
- 2. If found: use that record's competitor_id and core data
- 3. Run the deeper checks (PageSpeed, crawl, etc.) on top of saved data
- 4. If no record found: run fresh – both Lite and Full data – and store both

Example schema (Postgres / any DB):

```
leads
  lead_id (PK)
  business_name
  domain
  city, state
  vertical
  primary_keyword
  lead_gbp_rating, lead_review_count, lead_map_position
  competitor_name, competitor_domain, competitor_gbp_id
  competitor_rating, competitor_review_count, competitor_position
  lite_report_generated_at
  full_report_generated_at (nullable)
```

SECTION 4

Dynamic Content in the Full Report

The PDF reference (Empire_Audit_NEW_STRUCTURE.pdf) shows the visual format. But the **content of every page is dynamic** based on what the checker actually finds. Below is what's static (the structure) vs what's dynamic (the data) for each page.

Page 1 — Cover

Static: Layout, magazine-style header, "BY THE NUMBERS" and "INSIDE THIS REPORT" columns.

Dynamic: The big paradox headline. Choose from a templated set based on the lead's strongest paradox vs the competitor.

Headline templates (pick the strongest match):

- › More reviews, lower rank → *"Strong Reviews. Wrong Position. Bleeding Money."*
- › Higher rating, lower rank → *"Better Rated. Still Losing."*
- › Similar profile, lower rank → *"Same Story. Different Outcome."*
- › Lead not in top 10 → *"Invisible Where It Matters."*
- › No reviews vs competitor with reviews → *"Empty Profile. Empty Pipeline."*

Page 2 — Step 1 · Discovery

Static: Layout, map-pack visual, kicker "STEP 1 OF 4 · DISCOVERY."

Dynamic: Headline, subhead, all numbers in the map-pack visual, the math callout, and the 2–3 numbered fixes.

The fixes shown must mirror the **actual** issues found. If the lead's GBP completeness is 38%, fix #1 talks about completing it. If photos are missing, that's the fix. Don't show generic best-practice items.

Page 3 — Step 2 · First Impression

Static: Layout, comparison table format.

Dynamic: Every row in the comparison table reflects what was actually checked. Headline matches the most damaging finding.

Headline rule: The current "Three Seconds. No Reason to Stay." only works if speed/time-on-page is part of the finding. If the strongest issue here is, say, hidden phone number on mobile, headline becomes *"Phone Hidden. Customers Gone."* Always pick the headline that matches what the comparison table actually shows.

Comparison table rows are also dynamic. Show only the factors where the lead actually loses to the competitor. If the lead's CTA is fine but the trust badges are missing, show trust badges row, skip CTA row.

Fixes (numbered list below the table) must mirror table rows. If the table flags 3 issues, show 3 fixes — each addressing one of those rows. Not generic best-practice items.

Page 4 — Step 3 · Trust

Static: Layout, the two-column "You vs Competitor" numbers visual, kicker.

Dynamic: Everything inside — including the headline.

IMPORTANT — TRUST PAGE IS NOT ALWAYS THE REVIEWS PARADOX

The current reference shows "182 Reviews. Zero on Your Site." because that's Empire's specific situation. For a lead who *does* have reviews on their site but has fewer/worse reviews than the competitor, the page should show that gap instead. Possible angles:

- › Lead has 0 on site, has many on Google → reviews paradox (Empire case)
- › Lead has reviews on site but competitor has more → review count gap
- › Similar review counts but competitor has higher rating → rating gap
- › Lead doesn't respond to reviews, competitor does → response rate gap
- › Lead's reviews are old, competitor's are fresh → recency gap

Pick the angle where the lead loses most clearly. Fixes mirror the angle (e.g., for the response-rate angle, fixes are about replying to reviews and optimizing GBP).

Page 5 — Step 4 · Other Issues Found

Static: Page layout, kicker "STEP 4 OF 4 · OTHER ISSUES."

Dynamic: Headline, subhead, the issue blocks (1, 2, or 3), and the combined-impact callout.

This page is a **bucket** for additional issues the checker found that weren't already covered on pages 2–4. Possible issues include:

- › Domain mismatch (business name ≠ domain)
- › No service area pages for nearby neighborhoods/cities
- › No online booking / quote form
- › Slow mobile site speed
- › Missing financing / payment options
- › NAP inconsistency across listings
- › Missing local schema markup
- › No emergency / 24-7 phone messaging
- › Hours not clearly displayed
- › Forms ask for too many fields

Rendering rules:

- › Pick the **top 1–3 issues** the lead has that haven't been covered already.
- › Each issue gets one compact block (A, B, C) with title + body + impact line.
- › **If zero issues found, skip this page entirely.** The PDF becomes 5 pages instead of 6.
- › Combined impact callout at the bottom sums up estimated revenue at risk across the blocks shown.

Page 6 — Your Next Move (CTA)

Static: Layout, the two-card DIY / DFY structure, the big yellow CTA button.

Dynamic: The competitor name ("...how to overtake [Competitor]") and the booking link.

SECTION 5

How to Check Site Speed (Lead + Competitor)

Use the **Google PageSpeed Insights API**. It's free, official, and the same source Google uses for ranking signals — so the data carries weight when we show it to the lead.

Endpoint

```
GET https://www.googleapis.com/pagespeedonline/v5/runPagespeed
?url={URL_TO_TEST}
&strategy=mobile
&category=PERFORMANCE
```

Run it **twice per check**: once with strategy=mobile, once with strategy=desktop. Mobile is the priority — that's where most home-service search traffic comes from.

What to extract from the response

FIELD	PATH IN RESPONSE
Performance score (0–100)	lighthouseResult.categories.performance.score × 100
Largest Contentful Paint (LCP)	lighthouseResult.audits['largest-contentful-paint'].displayValue
Cumulative Layout Shift (CLS)	lighthouseResult.audits['cumulative-layout-shift'].displayValue
Interaction to Next Paint (INP)	loadingExperience.metrics.INTERACTION_TO_NEXT_PAINT_MS
Time to First Byte	loadingExperience.metrics.EXPERIMENTAL_TIME_TO_FIRST_BYTE

Critical rule — same check on both sites

USE THE SAME CODE PATH FOR LEAD + COMPETITOR

When we put a comparison in the report, the numbers must come from **identical methodology**. Run the exact same function with the exact same parameters for both URLs. Different methods = inconsistent numbers = lost credibility.

Rate limits + caching

Free tier is 25,000 queries per day. We won't hit that, but use a **cache layer** — if we've checked a domain in the last 7 days, reuse the result. Saves API calls and keeps the comparison numbers stable if Lite and Full are run days apart.

SECTION 6

Bonus Pay Breakdown

On top of the current scope of work, we are adding the following bonus pay for the new requirements in this brief:

DELIVERABLE	BONUS
Dynamic content in Full Report (Sections 4 + 5 of this brief). Every page's content adapts to what the checker actually found. Dynamic headlines, dynamic comparison table rows, conditional Step 4 page (skipped if no extra issues).	+\$100
Two-report system (Section 2 + 3 of this brief). Lite Report and Full Report from the same checker, with shared persistence so the same competitor and the same core numbers appear in both for the same lead — even when they're generated days apart.	+\$250
TOTAL BONUS	+\$350

Payment conditions

- › Bonus paid after both items are tested and verified on at least 3 different leads with different verticals and cities.
- › Lite and Full reports for the same lead must show the same competitor — verified by re-running the Full Report at least 3 days after the Lite ran.
- › Dynamic content must not produce a generic / template-feeling PDF. Empty fixes, hardcoded numbers, or competitor mismatches = no bonus.

Reference files attached:

- › *Empire_Audit_NEW_STRUCTURE.pdf* — the Full Report layout we're targeting. Treat as visual reference; content is dynamic per Sections 4 and 5.
- › This brief — read first, ask any clarifying questions before starting.